Model Cards and Human Review

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

TEACHER GUIDE · 1 OF 9

Model Cards and Human Review

> Prepare Model Cards and Human Review: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Write a **model card** whose claims match **evidence**

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min

> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval answer guidance

Proxy Features and Fairness

Without opening the old lesson, explain one key idea from Proxy Features and Fairness. Give an example and a mistake that the idea helps you avoid.

Look for: Identify possible proxy features

Reference example, not the only acceptable wording: Teacher explanation: X occurs 8/10 north and 2/10 south in this sample. This supports an association, not a deterministic identity rule or a causal explanation.

Improve Generalization

Without opening the old lesson, explain one key idea from Improve Generalization. Give an example and a mistake that the idea helps you avoid.

Look for: Compare **training** and validation **evidence** for model capacity

Reference example, not the only acceptable wording: Teacher explanation: Choose the line with validation MAE 3. The curve **training** MAE 0 does not compensate for validation MAE 8.

Overfitting

Without opening the old lesson, explain one key idea from Over**fitting**. Give an example and a mistake that the idea helps you avoid.

Look for: Compare **training** and validation **evidence** for model capacity

Reference example, not the only acceptable wording: Teacher explanation: Choose the line with validation MAE 3. The curve **training** MAE 0 does not compensate for validation MAE 8.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

> Draft five model-card headings for a classroom leaf classifier. Why must the card state that its labels are classroom categories rather than professional plant diagnosis?

Concept to explain

A **model card** defines **intended use**, **evidence**, limitations, risks, decision boundaries, monitoring, and human escalation.

Term	Meaning
model card	deployment-facing model documentation
intended use	approved purpose and setting
do-not-use	explicit prohibited application
human review	authorized person checks or decides

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow

> extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Teach two 45-minute sessions. This week focuses on **model card**s and review. Model the worked example first, collect the new W 4 case independently before feedback, and use the fresh retry to check transfer.

Award credit for traceable reasoning and bounded claims, including justified uncertainty.

Readiness answer

Data provenance, model version, full results, limitations and decision history.

Printed student lesson and workbook

Paper, pencils and a calculator for fractions

The fictional examples and a blank **evidence** table

Saved classroom project records for critique; no personal student dataset required

End goal for the pictorial case

By the end, I can explain **model card**s and **human review** using a traced example and a stated limitation.

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Explain the mechanism

> Model Model Cards and Human Review using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Trace the information, fitted behavior, and evaluation proof separately.

Question to explain

QUESTION Write a **model card** with a clear **do-not-use** case and review path.

MODEL IDEA A **model card** defines **intended use**, **evidence**, limitations, risks, decision boundaries, monitoring, and human escalation.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

How the mechanism works

UNDER THE HOOD **Intended use**: classroom exploration of visible leaf spotting. **Evidence**: test metrics, confusion matrix, subgroup slices, and data version.

Draw a fresh data → model → **prediction** → evaluation trace:

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

Step 1

STEP 1 **Intended use**: classroom exploration of visible leaf spotting.

Step 2

STEP 2 **Evidence**: test metrics, confusion matrix, subgroup slices, and data version.

Step 3

STEP 3 Do not use for pesticide, medical, or agricultural diagnosis.

Step 4

STEP 4 Uncertain, unfamiliar, or high-consequence cases route to a teacher or qualified adult.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

Concrete teaching case

The teaching case

A **model card** says healthy/spotted,**80%** accuracy only.

The teaching case

Add responsible-use **evidence**.

Exact explanation to model

Why the result follows

Intended users/conditions, labels/data, evaluation sets/metrics, class/subgroup results, versions, limitations, excluded uses and **human review**. Classroom classification is not professional diagnosis.

Explain the added worked example

Explain the added worked example · 1

A **model card** helps another person understand what was built and what **evidence** exists. It is a record, not a certificate of safety. Link statements to saved data and run records.

Use unknown or not evaluated where **evidence** is absent instead of filling gaps with optimistic language.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason.

Earlier weeks supplied the model-building **process**; this week examines **model card**s and review.

Explain the added worked example · 2

Example card: classroom parcel classifier v 1; purpose: compare toy classification methods on fictional clear/blocked labels. **Training** 40, validation 20, test 12 with separate record groups.

Frozen test TN 7 FP 1 FN 2 TP 2, blocked positive: accuracy 9/12, precision 2/3, recall 2/4. Always-clear baseline scores 8/12. No real parcel or student decision is supported.

Before continuing, underline the decision rule and circle the **evidence** that supports the conclusion. Explain the example to a partner without replacing the counts or conditions with a stronger claim.

Keep the distinction between an observed result and a proposed next step.

Key terms

Term	Meaning
Model card	A concise structured account of a model and its limits
Intended use	The purpose and conditions the evidence supports
Out-of-scope use	A purpose for which the model is not supported
Review status	The current state of checks and unresolved issues

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Model cards and review at a glance

PART	EVIDENCE OR ACTION	INTERPRETATION
Purpose and version	Toy classifier v 1	Fictional data only
Evidence	12 held-out cases	Full matrix and baseline
Limits and review	No real decisions	List unresolved coverage

The model card should name feature availability, preprocessing fitted on training, model settings, reference-label rules and test date. Include subgroup coverage if evaluated; otherwise say it was not assessed. Describe relevant failure examples without exposing personal information.

Verify the worked accuracy, precision and recall. Explain the steps or evidence behind your response.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?

Thinking move	Teacher prompt
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Model Cards and Human Review

Write a **model card** whose performance statement can be traced to a concrete evaluation.

The facts we will use

> **Supplied facts**
>
> Toy v = 3 classifier scored16/20 on five unseen plants with four views each, under indoor tablet lighting. Baseline scored14/20.

> **Supplied facts**
>
> All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/33

Model Cards and Human Review

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Name evidence

Record v = 3, 20 photo cases, 5 plant identities, group-separated split and indoor tablet conditions.



2. Compare baseline

v = 3 accuracy 80%; baseline70% on the same 20 cases.



3. State limits

Do not claim all cameras, outdoor conditions or disease diagnosis were validated.

CHANGE ONE THING

The publicity sentence becomes "80% accurate for every plant".

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Name evidence	Record v = 3, 20 photo cases, 5 plant identities, group-separated split and indoor tablet conditions.
2. Compare baseline	v = 3 accuracy 80%; baseline70% on the same 20 cases.
3. State limits	Do not claim all cameras, outdoor conditions or disease diagnosis were validated.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Read the picture in order. Do not claim all cameras, outdoor conditions or disease diagnosis were validated. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Keep the stated limit

A **model card** communicates **evidence**; it is not certification of safety, fairness or suitability for consequential plant decisions.

Change one thing in the pictured case

Change one thing

Change: The publicity sentence becomes “**80%** accurate for every plant”.

Predicted result

Prediction: Reject that unbounded statement.

Why the result follows

Reason: The measured rate belongs to twenty related indoor photos, not every plant or setting.

Ask while showing the picture

Ask while showing the picture · 3

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Model Cards and Human Review, reduce support deliberately and check individual reasoning before discussing answers.

Choose the pacing route

Compact alternative: this two-session sequence covers selected core tasks. Use the three-session weekly plan for the full lesson, independent transfer, feedback and project work; allow extra time for extensions.

Board sequence

1. Put the facts on the board

Write the given facts: A **model card** says healthy/spotted,**80%** accuracy only.

2. State the decision to explain

Write the question to explain: Add responsible-use **evidence**.

3. Reveal the worked reasoning

Reveal after **prediction**s: **Intended use**rs/conditions, labels/data, evaluation sets/metrics, class/subgroup results, versions, limitations, excluded uses and **human review**. Classroom classification is not professional diagnosis.

Hold back the independent task

Keep for an independent attempt: A **model card** says 90% accuracy but gives no **intended**

use, data conditions or human-review rule. What should be added before classroom use?

Classroom dialogue

Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Guided response: TN 7 TP 2 give 9/12 accuracy; TP 2/(TP 2+FP 1)=2/3 precision; TP 2/(TP 2+FN 2)=2/4 recall.

Exit response: Record a new version, threshold, reason, affected evaluations and **review status**; retain prior results and distinguish new independent **evidence** from reused development data.

Case-specific teaching sequence

Stage · Teacher action

Stage: Session 1 start to 7 minutes

Teacher action: Use the prerequisite question: What should remain visible after an evaluation? Check the explanation before proceeding.

Stage: 7 to 18 minutes

Teacher action: Read the opening and model the worked evidence. Ask students to identify model card in context.

Stage: 18 to 32 minutes

Teacher action: Complete W 1 and W 2 with support: draft a purpose; check arithmetic. Require the evidence, not only the final response.

Stage: 32 to 45 minutes

Teacher action: Read the visual row by row. Collect the guided explanation and compare reasoning in pairs. Connect each conclusion to the stated evidence.

Stage: Session 2 start to 8 minutes

Teacher action: Retrieve the distinction between model card and intended use. Invite a counterexample to an overconfident claim.

Stage: 8 to 20 minutes

Teacher action: Discuss the application and complete W 3: A card says fair and reliable but provides only accuracy. Request three additions.

Stage: 20 to 35 minutes

Teacher action: Collect W4 independently before giving hints. Assess these outcomes: Write a model card whose claims match evidence; Separate intended use from unsupported use. Record any support used.

Stage: 35 to 45 minutes

> **Teacher action:** Use W 5 and the exit response to assess limits. Give targeted feedback, then the fresh retry. Save evidence of independent understanding.

Reduce help deliberately

Completion practice

Completion task: Verify the worked accuracy, precision and recall.

Reasoning to listen for

Teacher check: Teacher explanation: TN 7 TP 2 give 9/12 accuracy; TP 2/(TP 2+FP 1)=2/3 precision; TP 2/(TP 2+FN 2)=2/4 recall.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for **Model card**s and review. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected repair

Expected correction: Ask for the relevant counts, units, procedure or saved record. For this week: Intended: compare toy classifiers on the documented fictional task.

Not supported: real parcel handling, student assessment or decisions outside the evaluated setup.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Write a bounded replacement for the publicity sentence. Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Reasoned teacher answer

“v = 3 matched16 of20 reference labels on five held-out plants photographed indoors by tablet; baseline matched14 of20 under the same conditions.” Changed-case check: Reject that unbounded statement.

The measured rate belongs to twenty related indoor photos, not every plant or setting.

Changed-case reasoning: Reject that unbounded statement. The measured rate belongs to twenty related indoor photos, not every plant or setting.

Collect each learner’s explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

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Run the weekly evidence lab

> Run the Model Cards and Human Review investigation and use the first unsupported step to choose a teaching response.

Keep the test conditions fixed, record every result, and change only one planned factor.

1. Write the hypothesis and expected **evidence** before running.

2. **Model Card** Clinic: compare two cards, find unsupported claims and missing gates, then author a release-ready version.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose and reteach

MISCONCEPTION TO REPAIR A disclaimer at the bottom of the screen is a complete safety plan.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

A final answer is enough: Ask for the relevant counts, units, procedure or saved record. For this week: Intended: compare toy classifiers on the documented fictional task.

Not supported: real parcel handling, student assessment or decisions outside the evaluated setup.

A stronger claim makes a stronger project: Record a new version, threshold, reason, affected evaluations and **review status**; retain prior results and distinguish new independent **evidence** from reused development data.

The new case can be copied from the example: The independent task changes the **evidence**. Require a fresh explanation: Request evaluated counts, split history, reference labels, baseline, errors and supported use.

Replace the universal claim with: performance scope is not established from the supplied **evidence**; the reported 90 percent cannot yet be audited.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Model Cards and Human Review questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Worked case answer

Teacher key · Worked case answer

Intended users/conditions, labels/data, evaluation sets/metrics, class/subgroup results, versions, limitations, excluded uses and **human review**. Classroom classification is not professional diagnosis.

Independent question

Independent question · withhold the key

A **model card** says 90% accuracy but gives no **intended use**, data conditions or human-review rule. What should be added before classroom use?

Expected independent answer

> **Teacher key · Expected independent answer**
>
> Add intended and excluded uses, dataset and evaluation conditions, subgroup results and limitations, plus when uncertain or consequential cases go to a human.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “One aggregate score is sufficient documentation for responsible use.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

> **Teacher key · Original transfer answer**
>
> Include **intended use**/users, data and label definitions, model/version, evaluation and subgroup results, limitations/excluded uses, and human-review **process**.

The card should connect claims to recorded tests and avoid presenting classroom labels as professional advice.

Explanations for the additional workbook examples

W1 Draft a purpose

Workbook question · W1

Write an intended-use and non-use statement for the worked card.

Teacher reasoning · W1

Teacher explanation: Intended: compare toy classifiers on the documented fictional task. Not supported: real parcel handling, student assessment or decisions outside the evaluated setup.

W2 Check arithmetic

Workbook question · W2

Verify the worked accuracy, precision and recall.

Teacher reasoning · W2

Teacher explanation: TN 7 TP 2 give 9/12 accuracy; TP 2/(TP 2+FP 1)=2/3 precision; TP 2/(TP 2+FN 2)=2/4 recall.

W3 Find missing evidence

Workbook question · W3

A card says fair and reliable but provides only accuracy. Request three additions.

Teacher reasoning · W3

Teacher explanation: Request full error counts and denominators, relevant subgroup coverage/results, data provenance and limits; remove unsupported fairness/reliability claims.

W4 Apply independently

Workbook question · W4

Review this card: model v 2, 90 percent accurate, works for every school. No dataset size, split or baseline is given. Write a correction request with a bounded replacement claim.

Teacher reasoning · W4

Teacher explanation: Request evaluated counts, split history, reference labels, baseline, errors and supported use.

Replace the universal claim with: performance scope is not established from the supplied **evidence**; the reported 90 percent cannot yet be audited.

W5 Maintain the record

Workbook question · W5

What should change in a card after a threshold revision?

Teacher reasoning · W5

Teacher explanation: Record a new version, threshold, reason, affected evaluations and **review status**; retain prior results and distinguish new independent **evidence** from reused

development data.

Assess independent transfer

Independent question · withhold the key

Selected case: Review this card: model v 2, 90 percent accurate, works for every school. No dataset size, split or baseline is given. Write a correction request with a bounded replacement claim.

Selected case answer

Teacher key · Selected case answer

Teacher explanation: Request evaluated counts, split history, reference labels, baseline, errors and supported use.

Replace the universal claim with: performance scope is not established from the supplied **evidence**; the reported 90 percent cannot yet be audited.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

> Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect and extend learning

> Connect Model Cards and Human Review to Responsible Plant Classifier, adapt support and plan a later changed-case check.

Responsible Plant Classifier

Complete the Plant Classifier **model card** and require teacher sign-off before any classroom demonstration.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Support and stretch

Use a tiny labelled dataset, explicit units and a named positive class. Stretch: require a baseline, held-out evaluation and subgroup or shifted-condition analysis with denominators.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or **human review**. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 34: Capstone Dataset and Baseline. Retrieve today's **evidence** and identify what carries forward and what is new.

Support and fresh reassessment

Provide a blank **evidence** table and allow a verbal explanation before writing. Highlight the given conditions without supplying the W 4 conclusion. Use **model card** and **intended use** in full sentences.

Extension: alter one condition in the worked example and explain which conclusion changes and which remains supported.

For independent W 4, require: Request evaluated counts, split history, reference labels, baseline, errors and supported use.

Replace the universal claim with: performance scope is not established from the supplied **evidence**; the reported 90 percent cannot yet be audited.

Fresh retry: a card lists test accuracy 18/20 but omits positive-class recall and **intended use**. Ask for full matrix, class definition and bounded use; do not infer recall from accuracy.

Mark each criterion as independent, supported or not yet demonstrated. Reassess the reasoning after feedback; do not replace the first attempt silently.

Next week: Capstone data and requirements. Carry forward the saved **evidence** and the distinction between development choices and justified claims.

Plan the delayed return

Suggested return points in this level: ai-4/34, ai-4/36. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Prepare, teach and assess

> Use this preparation summary alongside the detailed Model Cards and Human Review lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Write a **model card** whose claims match **evidence**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
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> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

Before class, solve the supported practice: Verify the worked accuracy, precision and recall.

Supported practice answer

Reasoned practice answer: Teacher explanation: TN 7 TP 2 give 9/12 accuracy; TP 2/(TP 2+FP 1)=2/3 precision; TP 2/(TP 2+FN 2)=2/4 recall.

Check every learner before discussion: Use the worked case for **Model card**s and review. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: Ask for the relevant counts, units, procedure or saved record. For this week: Intended: compare toy classifiers on the documented fictional task.

Not supported: real parcel handling, student assessment or decisions outside the evaluated setup.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: The publicity sentence becomes “**80%** accurate for every plant”. Require a **prediction** and a reason before comparing. Expected result: Reject that unbounded statement.

Reason: The measured rate belongs to twenty related indoor photos, not every plant or setting.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

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Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

For later coding lessons

> This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

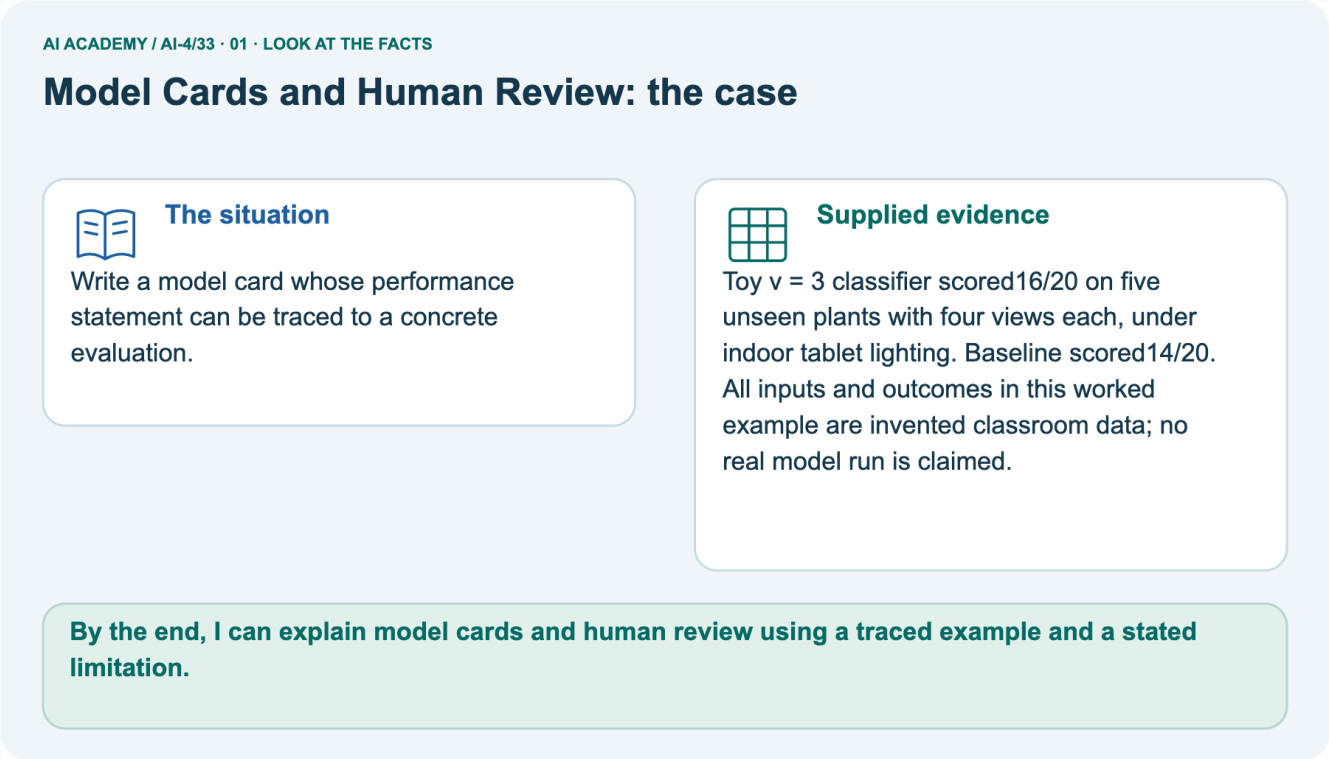
Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

Visual concept journey

> Show one Model Cards and Human Review graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Model Cards and Human Review: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

Write a **model card** whose performance statement can be traced to a concrete evaluation. Supplied facts: Toy v = 3 classifier scored16/20 on five unseen plants with four views each, under indoor tablet lighting.

Baseline scored14/20. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

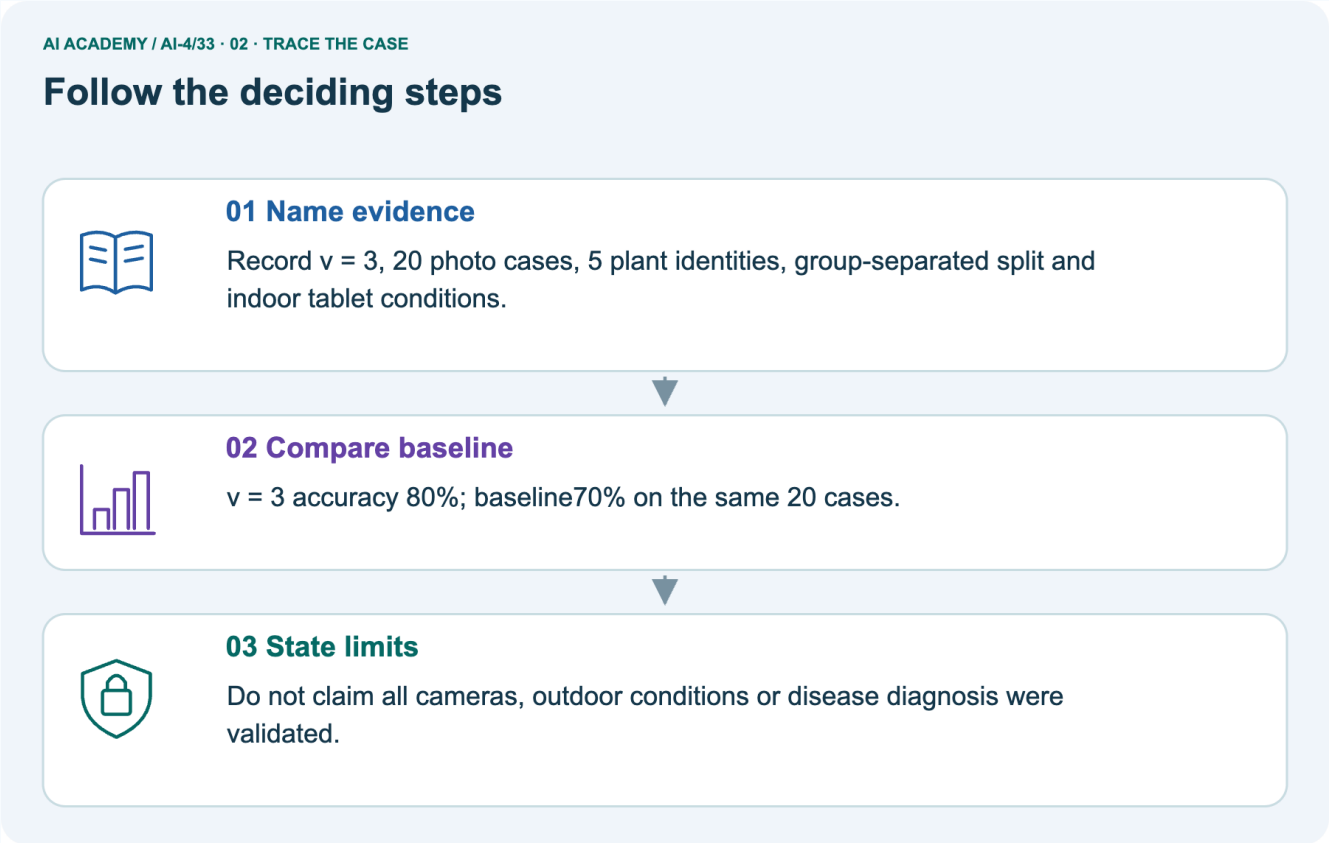
> Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the

image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Name **evidence**: Record v = 3, 20 photo cases, 5 plant identities, group-separated split and indoor tablet conditions. → Compare baseline: v = 3 accuracy **80%**; baseline**70%** on the same 20 cases. → State limits: Do not claim all cameras, outdoor conditions or disease diagnosis were validated.

Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Model cards and review at a glance



Follow the labelled picture with the matching explanation. Open the vector diagram

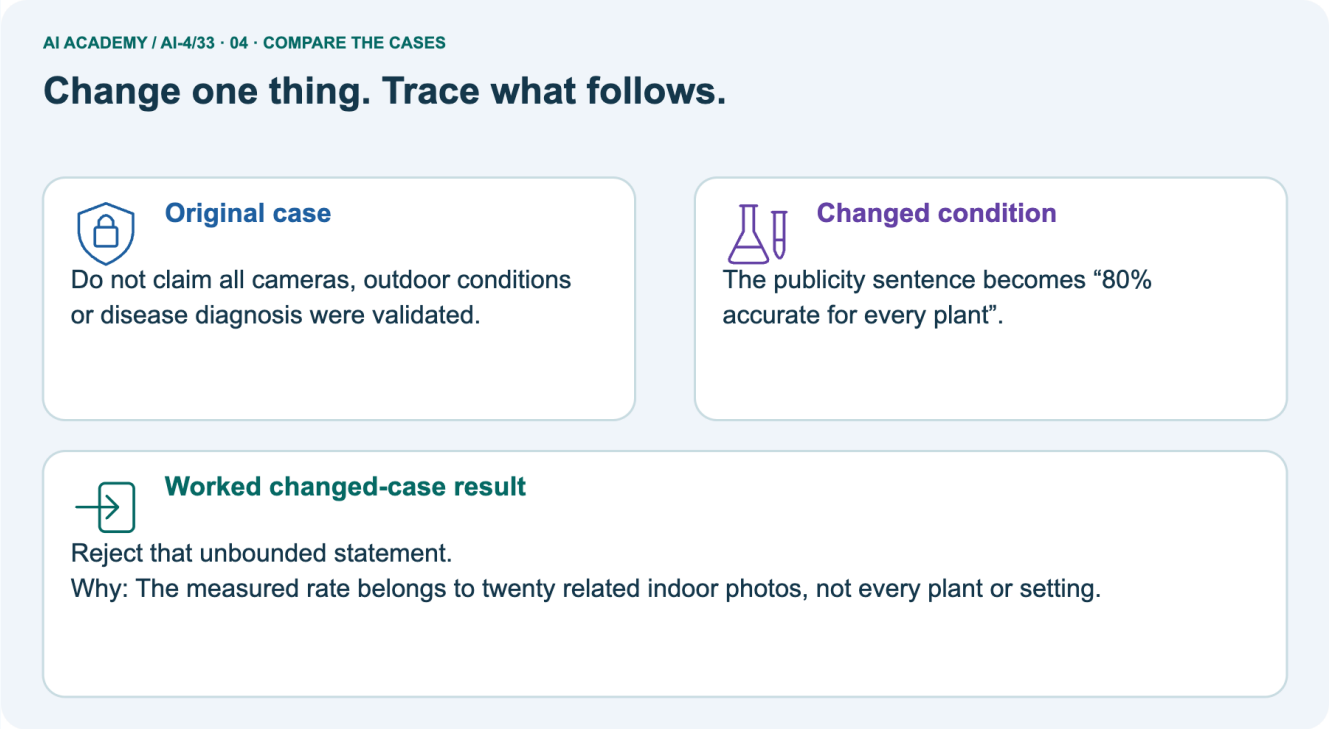
Model cards and review at a glance. Purpose and version | Toy classifier v 1 | Fictional data only. **Evidence** | 12 held-out cases | Full matrix and baseline. Limits and review | No real decisions | List unresolved coverage

Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Do not claim all cameras, outdoor conditions or disease diagnosis were validated. Changed condition: The publicity sentence becomes “**80%** accurate for every plant”.

Expected result: Reject that unbounded statement. Reason: The measured rate belongs to twenty related indoor photos, not every plant or setting.

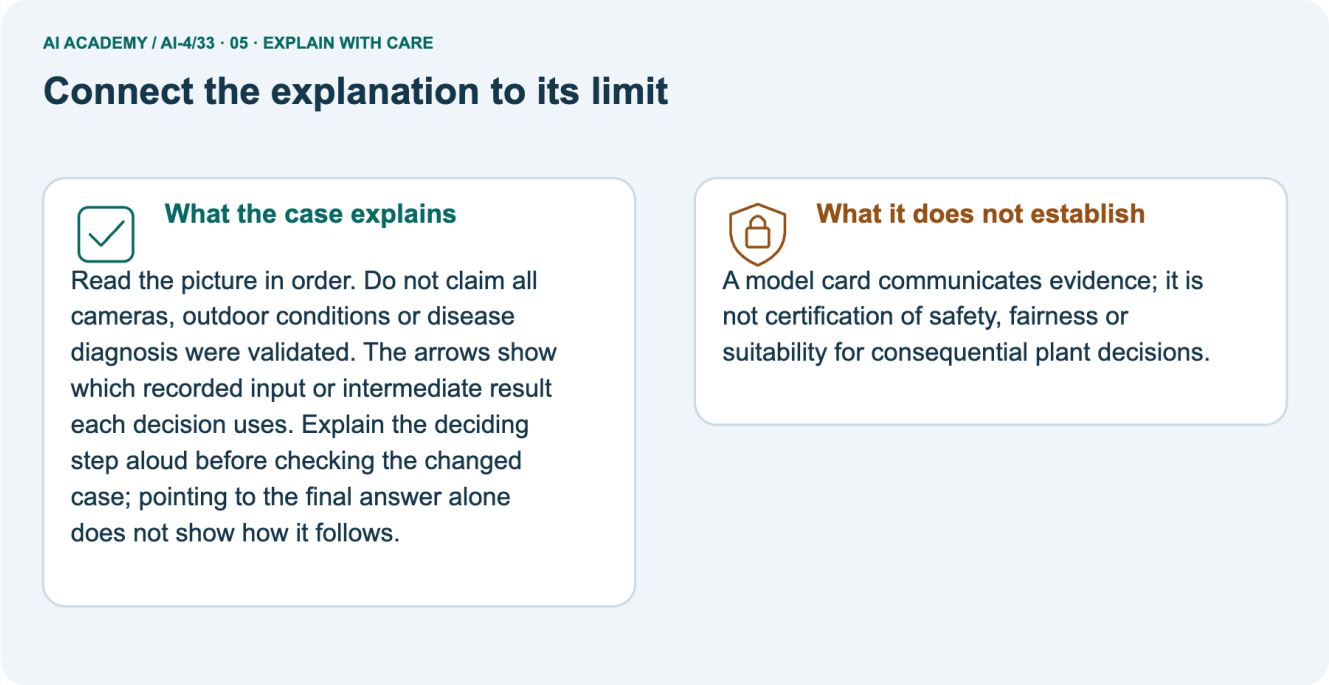
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The measured rate belongs to twenty related indoor photos, not every plant or setting.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Do not claim all cameras, outdoor conditions or disease diagnosis were validated. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: A **model card** communicates **evidence**; it is not certification of safety, fairness or suitability for consequential plant decisions.

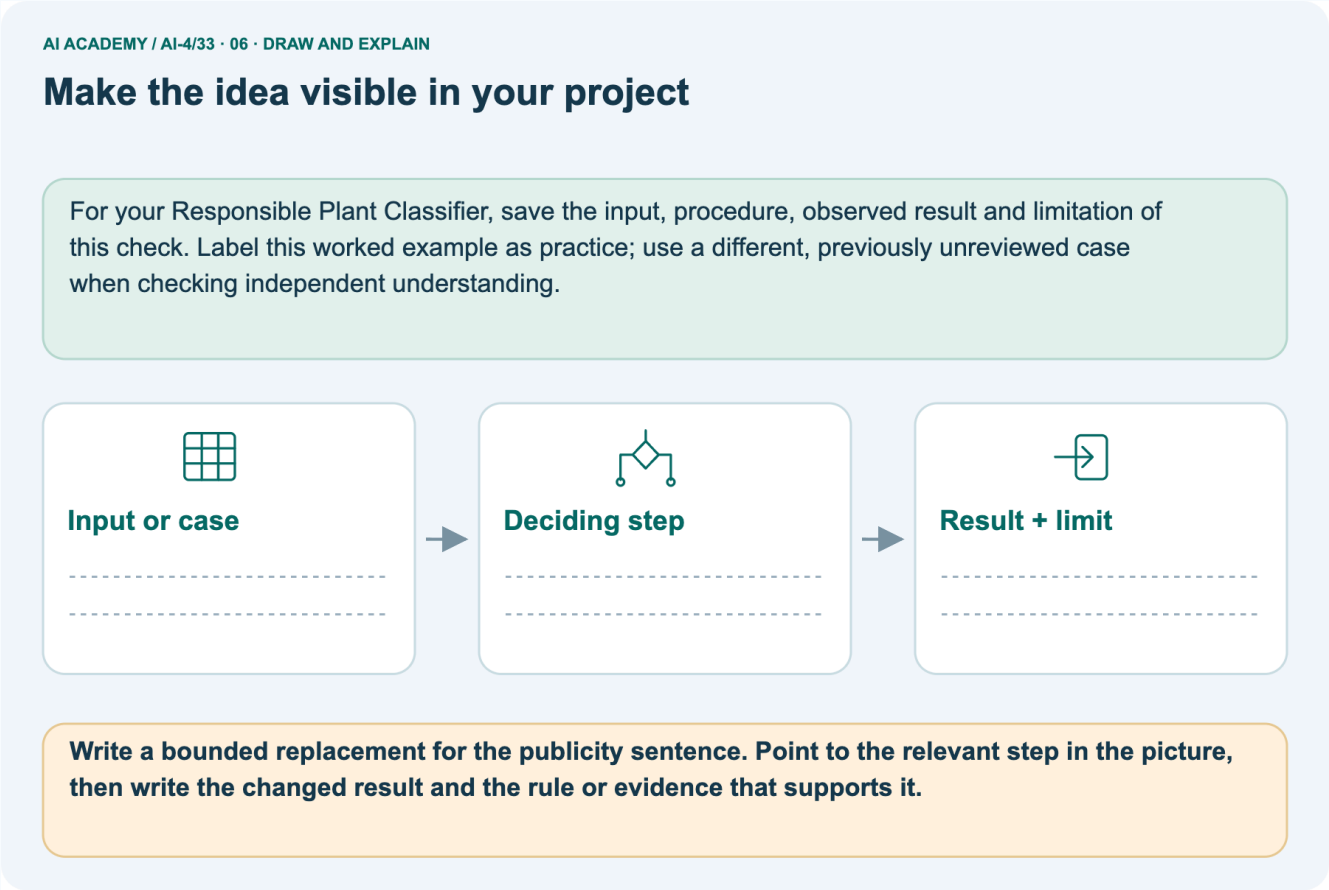
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: A **model card** communicates **evidence**; it is not certification of safety, fairness or suitability for consequential plant decisions.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Write a bounded replacement for the publicity sentence. Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: “v = 3 matched16 of20 reference labels on five held-out plants photographed indoors by tablet; baseline matched14 of20 under the same conditions.” Changed-case check: Reject that unbounded statement.

The measured rate belongs to twenty related indoor photos, not every plant or setting.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.